HadUK-Grid — How the Rainfall Figures Are Produced

Background note for the Lopcombe Corner analyses that use HadUK-Grid 1 km monthly rainfall (v1.3.2.ceda, release v20260512) · written 20 Oct 2026

In short. HadUK-Grid does not measure rainfall itself. It takes readings from the UK rain-gauge network and spreads them onto a regular 1 km grid by statistical regression and interpolation. The value for the Lopcombe Corner cell is therefore an **estimate** built from surrounding gauges and the local geography, not a reading from a gauge at the borehole.

1. Where the data come from

- The input is the Met Office archive of UK weather observations from land-based surface stations — a network of thousands of rain gauges whose membership changes over time as stations open and close.
- Monthly rainfall before 1961 was added through the **Rainfall Rescue** project, which digitised historical handwritten gauge records.
- Because every month is gridded afresh from whichever gauges were reporting, the grid smooths over the routine opening and closing of individual stations.

2. How the gauge readings become a grid

- **Regression on geography.** Gauges are irregularly placed, so the gaps are filled by a process of regression and interpolation onto a 1 km × 1 km grid. The regression accounts for latitude and longitude, altitude, terrain shape, coastal influence and urban land use.
- **Interpolation of what geography doesn't explain.** The Met Office cites a 2005 *International Journal of Climatology* paper for the monthly method. As we understand it, each month is treated as a departure from the long-term average (for rainfall, a percentage of normal), and the part not explained by geography is interpolated between nearby gauges with weights that fall off with distance. The Met Office methods page does not spell these details out, so they are our understanding of the published method rather than confirmed here.
- **Coarser products.** The 5, 12, 25 and 60 km grids are averages of the 1 km cells within each larger box. Country, administrative-region and river-basin series are averages of the 1 km cells within each area.
- **Long-term averages** (1961–1990, 1981–2010, 1991–2020) are calculated by averaging or summing the monthly or annual grids.

3. What this means for the Lopcombe Corner figures

Aspect	Implication
Nature of the value	A weighted estimate for the 1 km cell (centre E 425500 / N 135500) from nearby gauges, adjusted for terrain — not a direct measurement.
Independence from Broughton	Broughton (EA E13580) is very likely one of the gauges feeding the estimate, so HadUK-Grid and Broughton are not fully independent. Their close agreement (Oct–Dec r = 0.96) partly reflects this.
Strengths	No gaps; no jumps when stations close; quality-controlled and consistent from 1963 (and earlier). This is why it gave cleaner winter totals than summing daily Broughton data from the EA API, which had dropped days.
Cell choice	Neighbouring cells differ by at most 1.25 mm a month, so small errors in locating the borehole do not matter.
Limitations	Smooths out very local downpours; more uncertain in earlier decades and where gauges are sparse; monthly totals hide the timing of rain within a month.
Fitness for this work	Seasonal totals (Oct–Mar) drive the chalk aquifer, and these are what HadUK-Grid does best. The limitations matter little for the winter-peak and cyclicity analyses.

4. Data used in this project

- Product: HadUK-Grid gridded climate observations, 1 km, monthly rainfall, version v1.3.2.ceda, release v20260512, from the CEDA archive (`badc/ukmo-hadobs/data/insitu/MOHC/HadOBS/HadUK-Grid/`).
- Files: 63 yearly NetCDF files, 1963–2025, about 42 MB each, kept on the external SSD at /Volumes/Crucial X9 Pro for Mac/HadUK-Grid. Downloading needs a free CEDA account and an access token

(scripts/get_haduk_1km.sh handles this).

- Extracted series: data/haduk/lopcombe_haduk1km_monthly_rainfall.csv (cell value and 3×3 mean).

Sources

- Met Office, *HadUK-Grid Methods*: https://www.metoffice.gov.uk/research/climate/maps-and-data/data/haduk-grid/methods
- CEDA, *Using Archive Access Tokens*: https://help.ceda.ac.uk/article/5100-archive-access-tokens
- Monthly gridding method: 2005 *International Journal of Climatology* paper cited on the Met Office methods page; HadUK-Grid science paper also linked there. Not read directly for this note.